

Convolution in Tsetlin Machines

An interim report: what an agentic research team measured on CIFAR-10, and where it stopped

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Summary

A five-member agentic research team was asked to find the best way to handle convolution in Tsetlin machines and to close the gap to convolutional neural networks on CIFAR-10. It was halted after three days, part-way through its reproduction phase. This report states what was established, what was withheld, and what remains open.

Three results are solid. First, the number the field's CIFAR-10 progress is usually anchored to — **60.7%** — **is not a CIFAR-10 result**; it traces to an MSc thesis on a *modified* dataset, cited at third hand. The real published single-model bar is **75.4%**. Second, a **clause-size budget** — a published mechanism whose own authors measured it at **+0.06 points** on CIFAR-2 and presented as an interpretability and hardware device — is worth **+9.46 points** on convolutional CIFAR-10 at 4.8× fewer literals per image. It survives the exact sequential algorithm (+8.75) and it is *not* substitutable by the specificity parameter s . Third, the mechanism behind it is measured: literal inclusion is a **ratchet** whose admission rule never asks whether a literal is useful, so clauses accumulate logically redundant literals until they memorise patches rather than describe patterns.

Two central measurements are **withheld**. The capacity ladder is flat where the literature reports a +4.6 point rise over the same range, but at one seed and one fifth of the published epoch budget it cannot distinguish a reproduction failure from an under-trained arm. And the programme's own validation-selection rule was found to be a biased estimator carrying more noise than the decision threshold it feeds — a defect in our method, found by the team, affecting every arm, and fixable from stored data without a single re-run.

This is an *interim* report. The programme was halted part-way through its third phase; two of its central measurements are

incomplete and are reported as such. Every number in this document is generated from `experiments/convtm/results/*.json` by `code/report_data.py`. Nothing is hand-typed. 145 result records, 71 distinct arms, 10 decision records.

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1 Scope, and where this stopped

1.1 What was asked

Find the best way to handle convolution in Tsetlin machines; establish a CNN baseline on CIFAR-10 from the literature; re-implement the existing Tsetlin approaches and measure them under one harness; identify their limitations by measurement; design and test a new approach; and report. Two standing constraints governed the work: **the torchsetlin library was not to be modified** until explicitly released, and **no method was to be chosen without experimental results**.

1.2 Where it stopped

The programme completed its setup phase (P0), its literature phase (P1), and entered its reproduction phase (P3) with CNN baselines (P2) running concurrently. It was halted during P3. Consequently:

- **No new approach was designed or tested.** Phases P4 (limitation analysis), P5 (candidate generation) and P6 (evaluation) did not run. Nothing in this report claims a novel mechanism.
- **The reproduction phase is partial.** Four Tsetlin arms and eleven CNN baselines carry test numbers; the planned twelve-arm comparison did not complete.
- **The accuracy decomposition is incomplete.** Its two load-bearing controls — a CNN on the same Booleanized input, and a binarised CNN — were queued but had not run when work stopped.

1.3 What the constraints bought

Both held. `git diff - src/` is empty: not one line of the library was changed, and the eleven gaps found in it are recorded as reproductions with fix sketches rather than patches (§7). Every method decision in `DECISIONS.md` cites a measured result; where one had to be taken before its measurement existed, it was marked provisional and the measurement was scheduled before anything depended on it. Two decisions were subsequently overturned by their own falsifiers, which is the mechanism working rather than failing.

2 Method and protocol

2.1 Protocol

CIFAR-10, official train/test. A fixed stratified 45 000/5 000 train/validation split (`split_seed=1234`, hash `8a08ca15`) shared by every arm, Tsetlin and CNN alike. **Selection is on validation; the test set is evaluated once** per (arm, seed) at the selected epoch. This breaks comparability with the prior programme in `experiments/mctm`, which reported `best_test_acc`, i.e. test-set model selection; arms reused as baselines were re-run.

The seed band was measured, not assumed: five seeds of a 2 000-clause arm gave a standard deviation of 0.61 points, so **two three-seed means must differ by at least 1.00 points to count as a difference**. Validation selection contributes only about 0.09 of that.

2.2 Why the literature's numbers needed re-deriving

Three configuration facts had to be settled before any arm could be scored, and each was worth more than an experiment:

- **“Clauses” means *per class*** — confirmed in the source paper and independently in the reference implementation. The literature’s “2 000” is 20 000 total. Mis-reading this under-prices every arm by a factor of ten, and it is unqualified in two of the four papers that report CIFAR-10 numbers.
- **Load-bearing settings are undocumented.** A clause-size budget of 32 appears in all twenty-two reference scripts of the state-of-the-art toolbox and in *none* of its tables. Without it the headline cell does not reproduce. This produced a standing rule: *read the reference code before declaring any reproduction a failure.*
- **No Tsetlin image paper uses a validation split.** Reported accuracy is the mean of the last 25 or 100 epochs, or a peak. Our protocol therefore reports lower numbers for the same configuration — measured at ≤ 0.25 points, four times below the seed band.

2.3 What was built

A harness in `experiments/convtm/code/`: data pipeline and Booleanization registry, a result schema that *refuses* to write a record selected on test data, per-GPU job queues, and a diagnostics module recording per epoch the clause length, negation fraction, firing rate, per-clause match-count distribution, TA-state histogram and marginal selectivity by inclusion depth.

Two design decisions paid for themselves repeatedly. Storing the **full per-epoch curve** in every record means the selection defect of §4 can be repaired from stored data with no re-runs. And making rules **unstateable rather than merely written down** — a screening flag that enforces its own protocol, a validator that rejects test-selected records, a verdict generator that refuses to compare arms that are not budget-matched — caught three errors that discipline alone would not have.

3 Findings

3.1 The field's CIFAR-10 anchor is not a CIFAR-10 result

Progress in Tsetlin image classification is commonly measured against a convolutional Tsetlin machine result of **60.7%**. Tracing it: it originates in an MSc thesis (University of Agder, 2020) which, in the words of the paper that cites it, “utilized a *modified* CIFAR-10 dataset”. It is cited at third hand; its clause budget, window, T , s and epoch count are unknown. The 2019 Convolutional Tsetlin Machine paper reports **no CIFAR-10 number at all**.

The defensible bars are: best single model **75.4%** (64 000 clauses per class, 250 epochs); previous single-model state of the art **75.1%**; a vanilla convolutional machine **69.3%**; and **82.8%** for a *composite* of twenty-two specialists, which is an ensemble and is not comparable to a single model. This matters beyond bookkeeping: a programme aiming at 60.7% and one aiming at 75.4% are different programmes, and the gap to CNNs is 14.7 points smaller than the anchor implies.

3.2 Our harness reproduces the literature, and clears it

At 20 000 clauses, 5×5 colour thermometers, $T = 3000$, $s = 5.0$, budget 32, **50 epochs: 65.82% validation, 65.10% test**, against the source table’s **64.5%** at **250 epochs**. The reproduction target was cleared at one fifth of the training budget. This is the first time in this line of work that a published Tsetlin CIFAR-10 cell has been matched here.

3.3 The clause-size budget is worth nine points, not 0.06

A clause-size budget caps how many literals a clause may include. Its authors measured it at **+0.06** points on CIFAR-2 and **+0.05** on MNIST-with-convolution, and presented it as an interpretability and hardware mechanism. It is set to 32 in every reference script of the state-of-the-art toolbox and appears in none of that paper’s tables.

On convolutional CIFAR-10 we measure it at **+9.46 points** (47.77 ± 0.32 against 38.31 ± 1.05 , three seeds each, seed intervals disjoint), at **4.8× fewer included literals per image** and **+1.7% wall-clock**. For scale, the largest published single-model CIFAR-10 mechanism, Drop Clause, is worth +5.8.

s	feedback	unconstrained	sd	budget 32	sd	Δ
2	batch	36.78	0.99	48.31	0.17	11.53
10	batch	38.31	1.05	47.77	0.32	9.46
20	batch	36.39	0.41	45.60	0.49	9.21
10	sequential	38.06	0.38	46.80	0.17	8.75

Three things make this a finding about Tsetlin machines rather than about our implementation:

- **It survives the exact algorithm.** Under sequential feedback — the per-example algorithm, not our batched approximation — the effect is **+8.75**. The two modes differ by 0.71 points, inside the seed band, so batch aggregation is not producing it.
- **It is not the specificity parameter in disguise.** This was pre-registered as a rival explanation by the team member whose result it threatened, because s also shortens clauses and this repository’s own prior notes recommended using s instead. Tested: $s=2$ unconstrained reaches 36.78, against 47.77 for budget 32 — a gap of 11.15 points against a 1.00 point band. The budget pays at every s tested (+11.53 at $s=2$, +9.46 at $s=10$), and the best cell in the family is $s=2$ *with* the budget, at 48.31. The two dials are complementary.
- **It holds at scale.** At 20 000 clauses, a different encoding and a different s , the budget is still worth +6.43 points.

3.4 Why: inclusion is a ratchet

The mechanism was derived from the feedback rules and then measured. Adding a literal is effectively deterministic; removing one is not. An included literal can only be evicted when the clause *fails to fire*, at rate $1/s$; and the admission rule asks only whether a literal is *frequent inside the clause’s own match set*, never whether it is *useful*. The result is a ratchet.

Measured on an unconstrained clause over 30 epochs: clause length **86.7** $\rightarrow$ **170.6** (+97%, i.e. 68.7% of all includable literals) while the firing rate moves **−1.4%** and validation accuracy **+1.1** points. Epochs 5–30 alone add **34.4 literals for +0.32 points**, inside the seed band. **Clause length had not converged at 30 epochs; accuracy converged at epoch 4**. Per-literal selectivity is **4.6×** higher under the budget.

At 20 000 clauses the ratchet is not merely wasteful but harmful: the unconstrained arm’s validation accuracy **peaks at epoch 13 and declines by 2.30 points** by epoch 50 while its clauses grow to 188 literals. The machine trains itself backwards for 37 epochs. Only validation-selected reporting sees this; a last-epoch or last-25-epoch statistic — what every Tsetlin image paper in the bibliography uses — records the degraded number and never sees the peak.

One quantitative prediction of the theory was confirmed and one falsified. The predicted equilibrium firing rate $f^* = 1/(1 + s)$ holds to **11–13%** across a sevenfold movement in the predicted quantity. But clause length was predicted to fall below 32 at $s=2$ and instead stayed at 160: the admission threshold is *not* what governs clause length. What does govern it is open.

4 Withheld: the capacity ladder

The decisive experiment of the reproduction phase asked whether accuracy keeps rising with clause count. The literature says it does: the same encoding gains **+4.6** points from 20 000 to 80 000 total clauses and is still climbing at its largest measured cell. We measure it flat.

total clauses	our test	our val	plateau val	published (250 ep)	sel. epoch
20 000	65.1	65.82	64.2	64.5	45
40 000	65.44	66.2	64.55	66.8	34
80 000	63.61	65.58	63.91	69.1	16

The 20 000 point reproduces well. The curve then fails to rise: **+0.29 points on the plateau where the literature reports +4.6**.

This is reported as withheld, not as a result. Two explanations are live and one seed cannot separate them:

1. the published capacity rise does not reproduce; or
2. **50 epochs is too few at the top end.** The source ran 250. Our epoch correction was digitised from a figure at *one* clause count and may not transfer to a model four times larger. The convergence check is ambiguous: the 80 000 arm’s last ten validation epochs read 63.0, 63.4, 63.0, 64.9, 63.9, 62.8, 64.9, 64.8, 63.3, 63.3 — neither converged nor clearly climbing.

If (1) survives a second seed and a longer epoch budget it is a consequential negative result about a published table, and it would need stating very carefully. If (2) holds, the ladder simply needed more compute. The programme was halted before either could be run.

4.1 A defect in our own method, found by the team

While reading the ladder, the selection rule itself failed inspection. `selected_epoch = argmax(val_acc)` over 50 epochs is a **biased estimator on a noisy curve**. Measured across the three ladder arms: plateau standard deviation **0.80–0.94** points; binomial floor on 5 000 validation images at $p \approx 0.65$ is **0.67** points; and `argmax` inflates the reported validation figure by **+1.62, +1.65, +1.67** points — a consistency across three independent arms that is the signature of a pure order-statistic effect.

The consequence is that a decision rule with a 1.00 point threshold was being fed an estimator carrying about 1.65 points of selection noise, and the test accuracy at the winning epoch is an essentially uncorrelated draw from the plateau rather than the best model. The 80 000 arm’s selected epoch 16 is a noise spike; epochs 44–48 sit at the same level.

This is not leakage — selection remained validation-only throughout — but it is a precision defect,

and it is the reason the ladder verdict is withheld rather than merely under-seeded. The repair costs **no GPU time**: every record stores its full per-epoch curve, so every existing arm can be re-scored under a smoothed selection rule without a single re-run.

5 The CNN side, as far as it got

Eleven CNN baselines completed before the halt, all under the same split, the same validation-only selection and the same single test evaluation as the Tsetlin arms.

arm	test %	sd	seeds
cnn-resnet18-n1000	47.34	0.00	1
cnn-resnet18-n10000	85.90	0.00	1
cnn-resnet18-n5000	75.42	0.00	1
cnn-resnet20	91.70	0.00	1
cnn-resnet20-noaug	83.70	0.00	1
cnn-resnet56	92.59	0.00	1
cnn-small	92.71	0.00	1
cnn-small-noaug	89.13	0.00	1
cnn-small-noaug-n1000	53.59	0.00	1
cnn-small-noaug-n10000	79.16	0.00	1
cnn-small-noaug-n5000	72.25	0.00	1

The like-for-like comparison is against an *unaugmented* CNN, because the Tsetlin arms use no augmentation: **89.13%** for a simple convolutional network against **65.44%** for the best Tsetlin arm measured here — a gap of roughly **24 points**. Augmentation alone is worth 3.6 points to that network (92.71 against 89.13) and 8.0 to ResNet-20 (91.70 against 83.70), which is larger than most effects this programme is hunting and is the reason the unaugmented column is the honest one.

5.1 The decomposition that did not run

The report's intended spine was a decomposition of that 24-point gap into *Booleanization loss*, *binary-computation loss* and *Tsetlin-mechanism loss*, via two controls: a CNN trained on the same Boolean tensor the Tsetlin machine sees, and a binarised CNN. **Neither ran**. They were queued behind the arms above when work stopped.

What the literature already establishes is that the binary term is small: the best fully binary network (1-bit weights *and* activations) reaches 93.75% on CIFAR-10 against 94.8% for its own full-precision baseline, and binarising weights alone has been reported to *improve* accuracy. **So the gap is not explained by binary arithmetic**, and on the team's pre-registered estimate — recorded before the runs, and untested — roughly two thirds of it is architectural: one layer, no hierarchy, and a max/OR over patch positions rather than a learned pooled representation.

5.2 One comparison that needs no further runs

From capacity records already measured: the flagship Tsetlin arm holds **26.2 million automata = 104.6 MB of state**, which is **2.3× more storage than ResNet-18’s 11.2 million fp32 parameters**, and evaluates **436 million included literals per image**, between a simple CNN’s 165 million and ResNet-18’s 555 million multiply-accumulates — at 65.10% against ResNet-18’s expected mid-nineties.

The framing that Tsetlin machines are small and cheap does not survive contact with their own capacity records. Their advantages on this benchmark are interpretability and integer arithmetic, not size.

6 What did not work, and what we got wrong

6.1 Predictions that failed

Both specialists pre-registered falsifiable predictions on the round’s two hardest questions, and **both predictions failed by their authors’ own stated criteria**.

- **“Capacity saturates below 70% by 8–16k clauses.”** Falsified by the published table the author then read for themselves: the curve gains +2.6 points per doubling at exactly the point saturation was predicted. Withdrawn.
- **“A counting pool is worth 5–12 points.”** The supporting information-theoretic argument was recomputed from measured quantiles and found to be wrong by a factor of 5.7 — the original figure was a ceiling under a uniform distribution quoted as an estimate. Revised to +0–1.5 points, at or below the seed band, and withdrawn as a standalone candidate.
- **“The clause-size effect is an artefact of batched feedback.”** Falsified: the effect survives sequential feedback almost unchanged. The author’s diagnosis of their own failure is worth recording — the direction was right and the magnitude wrong by a factor of about 20.
- **“Clause length is governed by the admission threshold $1/(1 + s)$.”** Falsified: a tenfold change in s moves clause length by 7.7%. The companion prediction sharing the same constant, the equilibrium firing rate, was confirmed to within 11–13%.

6.2 Two corrections to the falsification method itself

Both came from the members they convicted.

A falsifier stated over a quantity the intervention itself changes is not a falsifier. Both of the first round’s pre-registrations were stated over a clause-match distribution that is the equilibrium of a model trained *under the very pooling rule the candidate replaces*.

A pre-registration must name at least one rival explanation and the measurement that separates it, not only the falsifier of the preferred one. The prediction that the clause-size effect was a batching artefact was correctly aimed by the first rule and still mis-aimed, because it named only the mechanism its author had thought of and never registered the cheaper rival — an untuned specificity parameter — which was the explanation that actually had to be excluded.

6.3 Errors in the conduct of the work

Recorded because a report that lists only what went right is not evidence of anything.

- **The team’s own rulebook was wrong and cost a day.** It stated, on the strength of an earlier measurement at budget 8, that clause-size control “does not bind”. It binds at every budget a published recipe uses. Three agents were steered away from what turned out to be the largest measured effect in the programme.
- **A signal sat unread in the data.** The diagnostic showing 156-literal clauses was in every record from the first run and printed on every epoch line; it was logged as a curiosity. The engineer’s own summary: *being right in a JSON nobody acts on is the same as being wrong*. An escalation threshold was added.
- **Three file edits silently did nothing** — a string replacement that no-ops on a missed target while reporting success. Two were still broken when discovered. A success message that is not evidence of success recurred three times in different guises (that one, a queue log that cannot distinguish “working” from “all jobs crashed”, and a fail-soft handler that would have written an invalid record after hours of training).
- **A hyperparameter was inert in the baseline.** T exceeded the achievable class sum, so the margin never annealed. Correcting it *increased* the headline effect by 2 points.
- **The orchestrator misdiagnosed a running queue as a crash** and had to retract it before it entered the audit log; and separately caused GPU contention that inflated one arm’s timings by authorising a job against a stale idle check.
- **A HOG encoder was wrong in every parameter that mattered** and was caught only when the correct specification arrived. It was recoverable because it had been registered as an explicit ambiguity, so the harness was already refusing to score it.

7 Proposed torchsetlin changes — not applied

The library was not modified. Eleven gaps were recorded, each with a minimal reproduction, an impact statement and a fix sketch. The ones that affected this work:

- **The documented claim that `max_included_literals` does not bind is too strong.** It overshoots by about +2.5 literals at budget 8 and *holds* at 16, 32 and 64 at batch 50. Given §3.3, this is the most consequential documentation error found: the repository’s `CLAUDE.md` and its stored notes still state the strong form, and it steers users away from a nine-point mechanism. **Recommended first change.**
- **The default chunk budget silently serialises large convolutional models** — a batch of 50 runs as one or two chunks at large clause counts on a card with 23 GB free, costing 1.2–1.5× throughput. Accuracy-neutral (verified), so it is a dial, not a bug, but the default is wrong for this regime.
- **No public API reaches the patch-match tensor.** Three separate diagnostics and every pooling or multi-layer candidate go through private hooks. Sketch: expose `clause_maps(x, include=None)`.
- **Memory scales with clauses × literals, not clauses**, through the feedback accumulator, and nothing warns about it. A wide-feature arm exhausts a 24 GB card at a clause count a narrow one runs comfortably.
- **A convolutional `_encode` override cannot wrap the parent’s**, only replace it.

- **No dilation**, and one patch size per model — the two cheapest receptive-field mechanisms, one of which is a single unthreaded keyword argument.
- **Booleanization registry lacks Otsu, Canny, HOG and adaptive multilevel colour thermometers** — four of the seven encodings behind the field’s best published result.

A further warning worth carrying into any future change: **do not offer float16 as a memory option**. Clause-violation counts sum up to the literal count — 11 664 for one encoding used here — and float16 represents integers exactly only to 2 048, so it would silently miscount on exactly the models large enough to want it.

8 What would settle the open questions

In cost order, on the two GPUs used here.

1. **Re-score every record under a smoothed selection rule**. No GPU time. Repairs the defect of §4 from stored curves and tightens every comparison in the programme. Must be done before anything else is concluded.
2. **Finish the accuracy decomposition** — the Booleanized-input CNN and the binarised CNN, roughly 18 GPU-hours. These are the two controls that turn “Tsetlin machines trail CNNs by 24 points” into a statement about *which* component is responsible, and neither exists in the literature.
3. **Settle the capacity ladder**: a second seed and a longer-epoch run at 80 000 clauses, roughly 15 GPU-hours. This decides whether a published table reproduces.
4. **The bloat-versus-headroom discriminator**, 0.6 GPU-hours: run the clause-budget pair on CIFAR-2 built from this pipeline. It separates “the effect tracks how bloated the baseline is” from “the effect tracks how much headroom the task has”, which is the difference between a general claim and a CIFAR-10 one.
5. **The optimum budget and what governs clause length**, roughly 5 GPU-hours. The budget sweep was prepared and not run; and the falsification of the admission-threshold account leaves the governing quantity unidentified, which is the most interesting open theoretical question the programme produced.
6. **The diversity-versus-ensembling control**, 3 GPU-hours. Twenty-two small specialists beat one large model by 4.1 points at 0.69× the automata; whether that is *encoding diversity* or merely *ensembling* has never been tested, and it is the cheapest route to a contribution this programme identified.

Only after (1)–(3) would candidate generation for a new mechanism be well-founded. The team’s own pre-registered reading — untested — is that the gap is predominantly architectural rather than a property of the learning rule, which would direct candidate generation at hierarchy, pooling that preserves count information, and receptive-field growth, and away from modifications of the feedback rule.

9 Reproducibility

Everything is under `experiments/convtm/`. 145 result records across 71 distinct arms; 10 decision records in `DECISIONS.md`, append-only, each citing the measurements it rests on; the round's deliberation in `rounds/ROUND-1/`; the audit trail in `AUDIT.md`.

Every record carries its command line, seed, git revision, environment, split hash, hyperparameters, capacity, full per-epoch curve and diagnostics, and refuses to be written if it was selected on test data. Predictions are stored per arm so paired significance tests need no re-runs.

Two verification results are worth stating. **The harness reproduces bit-exactly across a five-file change:** a re-run under new labels returned 37.50/37.93/39.50 and 47.80/48.07 — identical to the digit — which is what licenses treating the diagnostics added mid-programme as non-perturbing. And **cross-device runs agree within the seed band**, so `device + seed`, not `seed`, is the reproducibility key.

`git diff - src/` is empty. The library was never modified.

Status

This programme was halted part-way through its reproduction phase. It did not design or test a new approach to convolution in Tsetlin machines. What it establishes is a corrected set of reference numbers, a reproduced published cell, one mechanism whose published value understates its effect on CIFAR-10 by two orders of magnitude, an account of why, and an instrumented harness in which the remaining questions are a few tens of GPU-hours apart rather than open-ended.